Signal Protocol-WhatsApp using PQC

Sreejit Maity

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Introduction

To understand how modern messaging apps like WhatsApp and Signal work using the Double Ratchet protocol, and how to replace standard keys with Post-Quantum Cryptography (PQC) so future quantum computers cannot read past chats.

Key Concepts

- **Signal Protocol:** Creates a new shared key for every single message. If one message gets intercepted, past and future messages stay safe.
- **Why Classical Crypto Fails:** Signal normally uses Elliptic Curve Diffie-Hellman (ECDH). Shor's algorithm on a quantum computer can break ECDH easily.
- **Post-Quantum Fix:** We add a lattice-based Key Encapsulation Mechanism (KEM) like ML-KEM (Kyber) into the key exchange. Both parties mix the classical key and the quantum key together.

Step-by-Step Working

1. **Handshake (Initial Setup):** Alice and Bob exchange their public keys. Alice encapsulates a shared secret using Bob's post-quantum public key.
2. **Hybrid Secret derivation:** Both derive a root key by combining the classical secret and the quantum secret using a Key Derivation Function (KDF).
3. **Message Ratchet:** For every outgoing message:
    - Derive a message encryption key from the chain key.
    - Update the chain key using a hash function (like SHA-256).
    - Encrypt the text using AES or ChaCha20.

Simplified C Implementation

The following C code demonstrates the hybrid key mixing and the continuous message ratcheting mechanism.

```c
#include <stdio.h>
#include <string.h>

// Simple toy hash function to act as KDF for demonstration
unsigned int toy_hash(unsigned int input) {
    return (input * 2654435761u) ^ 0x5bf03635;
```

```
}

int main() {
    printf("--- Quantum-Safe Ratchet Demo ---\n");

    // Step 1: Hybrid Shared Secret
    unsigned int classical_shared = 0x1234;
    unsigned int pqc_shared = 0xABCD;

    // Mix both secrets together
    unsigned int root_key = toy_hash(classical_shared ^ pqc_shared);
    printf("Root Key initialized: 0x%X\n\n", root_key);

    // Step 2: Message Ratchet loop
    unsigned int chain_key = root_key;
    char *messages[3] = {"Hello", "How are you?", "Quantum secure text"};

    for (int i = 0; i < 3; i++) {
        // Derive message key and advance the chain
        unsigned int message_key = toy_hash(chain_key + 1);
        chain_key = toy_hash(chain_key);

        printf("Message %d: \"%s\"\n", i + 1, messages[i]);
        printf("  -> Message Key : 0x%X\n", message_key);
        printf("  -> Next Chain Key: 0x%X\n", chain_key);
    }

    return 0;
}
```

Conclusion

Even if a quantum computer breaks classical keys in the future, messages recorded today cannot be decrypted because the root key requires breaking the lattice problem as well.